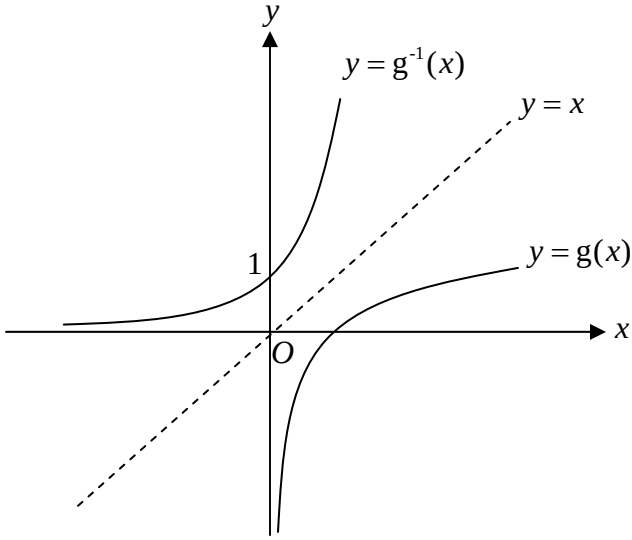


2008 H2 MATHS (9740) JC 2 PRELIM EXAM P1 – SOLUTIONS

Qn	Solution
1	Functions
(i)	
(ii)	$R_g = (-\infty, \infty), \quad D_f = (0, \infty)$ For fg to exist, $R_g \subseteq D_f$, largest domain of g is $D_g = (1, \infty)$. $D_g = (1, \infty) \rightarrow R_g = (0, \infty) = D_f \rightarrow R_{fg} = [-1, 1]$ $\therefore R_{fg} = [-1, 1]$

Qn	Solution
2	Complex Number 1
(a)	$iw + 3z = 2 + 4i \text{ -----(1)}$ $w + (1-i)z = 2 - i \text{ -----(2)}$ From (2), multiply by i, $iw + (i+1)z = 2i + 1 \text{ -----(3)}$ (1)-(3): $(iw + 3z) - (iw + iz + z) = (2 + 4i) - (2i + 1)$ $2z - iz = 1 + 2i$ $z(2-i) = 1 + 2i$ $z = \frac{1+2i}{2-i} = i$ Sub $z = i$ into (2), $w + (1-i)i = 2 - i$ $w = (2-i) - (i+1)$ $w = 1 - 2i$ $\therefore z = i, w = 1 - 2i$
(b)	$z = \frac{1-ki}{\sqrt{3}+i}$ $z = \left(\frac{1-ki}{\sqrt{3}+i} \right) \left(\frac{\sqrt{3}-i}{\sqrt{3}-i} \right)$ $z = \left(\frac{\sqrt{3}-i-k\sqrt{3}i+ki^2}{(\sqrt{3})^2-i^2} \right)$ $z = \left(\frac{\sqrt{3}-k-i-k\sqrt{3}i}{4} \right)$ For z to be purely imaginary, $k = \sqrt{3}$. <u>Alternative Method</u> Let $\frac{1-ki}{\sqrt{3}+i} = \alpha i$. $\Rightarrow 1-ki = \alpha\sqrt{3}i - \alpha$ $\therefore \alpha = -1, \quad k = -\alpha\sqrt{3} = \sqrt{3}$

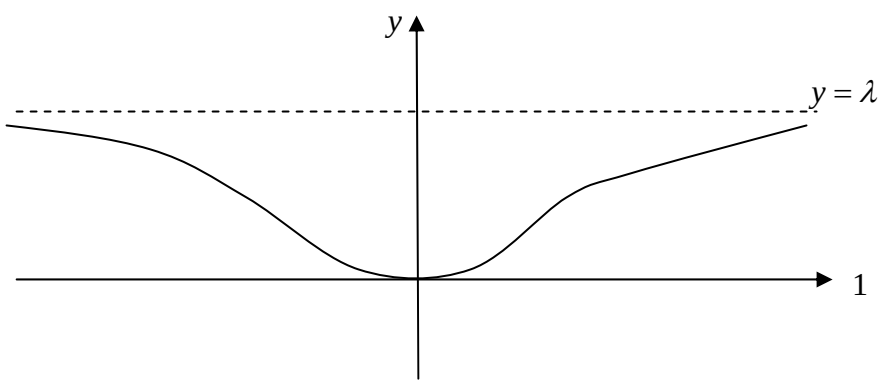
Qn	Solution
3	Integration Techniques
	$\int (x \ln x)^2 dx$ $= \frac{x^3}{3} (\ln x)^2 - \int \frac{x^3}{3} \left(2 \ln x \frac{1}{x} \right) dx$ $= \frac{x^3}{3} (\ln x)^2 - \frac{2}{3} \int (x^2 \ln x) dx$ $= \frac{x^3}{3} (\ln x)^2 - \frac{2}{3} \left[\frac{x^3}{3} \ln x - \int \left(\frac{x^3}{3} \frac{1}{x} \right) dx \right]$ $= \frac{x^3}{3} (\ln x)^2 - \frac{2x^3}{9} \ln x + \frac{2x^3}{27} + C$ $\int \frac{x+1}{1+2x^2} dx$ $= \frac{1}{4} \int \frac{4x}{1+2x^2} dx + \int \frac{1}{1+2x^2} dx$ $= \frac{1}{4} \ln(1+2x^2) + \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2}x) + C$

Qn	Solution
4	Mathematical Induction
	Let P_n be the statement “$\sum_{r=1}^n (2r-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$” for $n \in \mathbb{N}^+$ For $n=1$, $\text{LHS} = 1^2 = 1$ $\text{RHS} = \frac{1}{3}(2-1)(2+1) = 1$ i.e LHS = RHS $\Rightarrow P_1$ is true. Assume that P_k is true for some $k \in \mathbb{N}^+$, i.e $\sum_{r=1}^k (2r-1)^2 = \frac{1}{3}k(2k-1)(2k+1)$. Need to show P_{k+1} is true, i.e $\sum_{r=1}^{k+1} (2r-1)^2 = \frac{1}{3}(k+1)(2k+1)(2k+3)$. For $n = k+1$, $\text{LHS} = \sum_{r=1}^{k+1} (2r-1)^2 = \sum_{r=1}^k (2r-1)^2 + (2(k+1)-1)^2$ $= \frac{1}{3}k(2k-1)(2k+1) + (2k+1)^2$

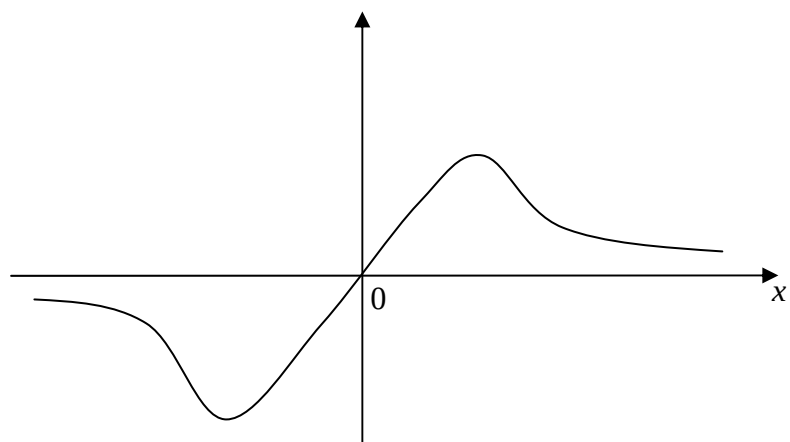
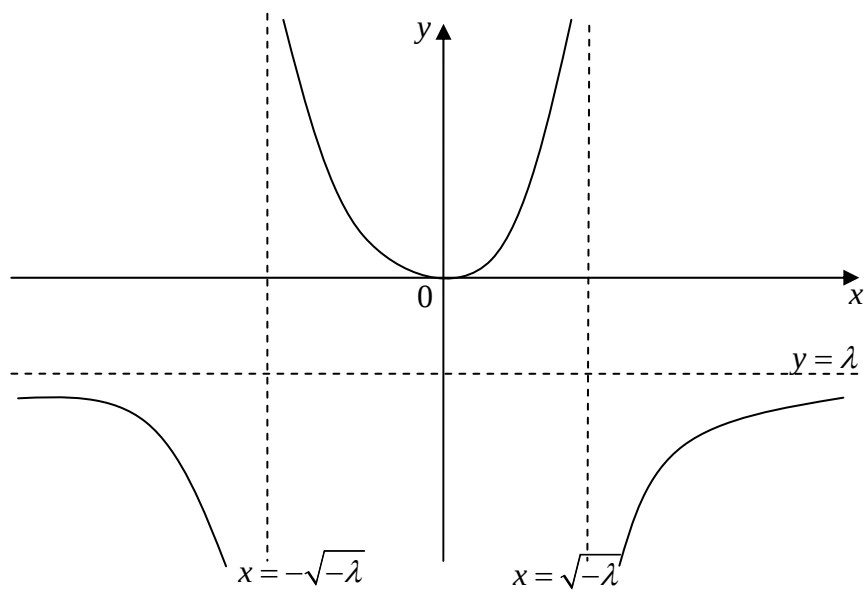
$$\begin{aligned}
&= \frac{1}{3}(2k+1)[2k^2 - k + 3(2k+1)] \\
&= \frac{1}{3}(2k+1)[2k^2 + 5k + 3] \\
&= \frac{1}{3}(2k+1)(2k+3)(k+1) = \text{RHS}
\end{aligned}$$

Thus P_k is true $\Rightarrow P_{k+1}$ is also true.

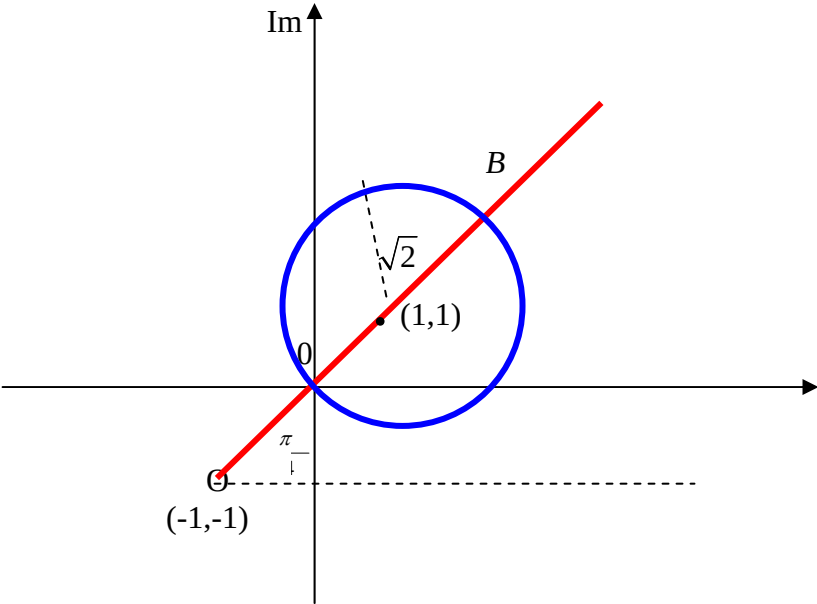
Therefore since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is also true, then by mathematical induction, P_n is true $\forall n \in \mathbb{N}^+$.

Qn	Solution
5	Curve Sketching & Transformation
(i)	

(ii)



Qn	Solution
6	Vectors 1 & 2
(a)	$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ $\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$ $ \mathbf{a} \mathbf{b} \cos \theta = a_1 b_1 + a_2 b_2 + a_3 b_3$ $\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2} \cos \theta = a_1 b_1 + a_2 b_2 + a_3 b_3$ $(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) \cos^2 \theta = (a_1 b_1 + a_2 b_2 + a_3 b_3)^2$ $\cos^2 \theta = \frac{(a_1 b_1 + a_2 b_2 + a_3 b_3)^2}{(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)} \leq 1$ $\therefore (a_1 b_1 + a_2 b_2 + a_3 b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$ <u>Alternative Method</u> $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \mathbf{b} \cos \theta$ $\Rightarrow \mathbf{a} \cdot \mathbf{b} \leq \mathbf{a} \mathbf{b} \quad \because \cos \theta \leq 1$ $\Rightarrow \mathbf{a} \cdot \mathbf{b} ^2 \leq \mathbf{a} ^2 \mathbf{b} ^2$ If $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix},$ $\Rightarrow (a_1 b_1 + a_2 b_2 + a_3 b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$
(b)	$\mathbf{p} \times 3\mathbf{q} = 2\mathbf{p} \times \mathbf{r}$ $\Rightarrow \mathbf{p} \times 3\mathbf{q} - 2\mathbf{p} \times \mathbf{r} = \mathbf{0}$ $\Rightarrow \mathbf{p} \times (3\mathbf{q} - 2\mathbf{r}) = \mathbf{0}$ $\Rightarrow \mathbf{p} \parallel 3\mathbf{q} - 2\mathbf{r}$ $\Rightarrow \mathbf{p} = k(3\mathbf{q} - 2\mathbf{r}), \quad k \in \mathbb{R}$

Qn	Solution
7	Complex Numbers 2
(a)	(i) $z - 1 - i = 1 - i = \sqrt{2} \Rightarrow z - (1 + i) = \sqrt{2}$ (ii) $\arg(iz - 1 + i) = \frac{3\pi}{4} \Rightarrow \arg(i(z + 1 + i)) = \frac{3\pi}{4} \Rightarrow \arg(i) + \arg(z + 1 + i) = \frac{3\pi}{4}$ $\therefore \frac{\pi}{2} + \arg(z + 1 + i) = \frac{3\pi}{4} \Rightarrow \arg(z + 1 + i) = \frac{\pi}{4} \Rightarrow \arg(z - (-1 - i)) = \frac{\pi}{4}$  From the diagram, the complex number satisfying both (i) and (ii) are: By similar triangles: $z = 0 + 0i = 0$ and $z = 2 + 2i$.

Qn	Solution
8(a)	Recurrence relations
(i)	$x_n = 0.9x_{n-1} + 90, \quad n = 1, 2, 3, \dots$
(ii)	$ \begin{aligned} x_n &= 0.9x_{n-1} + 90 \\ &= 0.9(0.9x_{n-2} + 90) + 90 \\ &= 0.9^2 x_{n-2} + 0.9(90) + 90 \\ &= 0.9^2 (0.9x_{n-3} + 90) + 0.9(90) + 90 \\ &= 0.9^3 x_{n-3} + 0.9^2(90) + 0.9(90) + 90 \\ &= \dots \\ &= 0.9^n x_0 + 0.9^{n-1}(90) + \dots + 0.9^2(90) + 0.9(90) + 90 \\ &= 0.9^n x_0 + 90 \left(\frac{1 - 0.9^n}{1 - 0.9} \right) \\ &= 0.9^n x_0 + 900(1 - 0.9^n) \\ &= 0.9^n (x_0 - 900) + 900(\text{shown}) \end{aligned} $
(iii)	As $n \rightarrow \infty, 0.9^n \rightarrow 0 \therefore x_n \rightarrow 900$
(b)	Let the sequences tend to α. Thus $\alpha = 0.3\alpha + 7 \Rightarrow \alpha = 10$ Also $\alpha = \sqrt{\frac{100}{\alpha}} + a$ $\therefore 10 = \sqrt{(10) + a}$ $a = 90$

Qn	Solution
9	Complex Numbers 3
	(i) $(z + 2i)^6 = z^6$ has only five roots since the coefficient of z^6 is one on both sides of the equation and the resulting polynomial is one of degree 5 (ii) $(z + 2i)^6 = z^6 \Rightarrow \left(\frac{z + 2i}{z}\right)^6 = 1 \Rightarrow \frac{z + 2i}{z} = e^{i\frac{2k\pi}{6}} = e^{i\frac{k\pi}{3}}, k = \pm 1, \pm 2, 3$ $\Rightarrow z + 2i = ze^{i\frac{k\pi}{3}} \Rightarrow z = \frac{2i}{e^{i\frac{k\pi}{3}} - 1}, \text{ where } k = \pm 1, \pm 2, 3 \text{ shown}$ (iii) Note: $z = \frac{2i}{e^{i\frac{k\pi}{3}} - 1} = \frac{2ie^{-\frac{k\pi}{6}i}}{e^{i\frac{k\pi}{6}} - e^{-i\frac{k\pi}{6}}} = \frac{2i\left(\cos\frac{k\pi}{6} - i\sin\frac{k\pi}{6}\right)}{2i\sin\frac{k\pi}{6}}$ $= \cot\frac{k\pi}{6} - i$ Thus the roots of the equation are: $z = \sqrt{3} - i, -\sqrt{3} - i, -\frac{\sqrt{3}}{3} - i, \frac{\sqrt{3}}{3} - i, -i$

Qn	Solution
10	Integration Applications
(a)	area of $R = \int_0^1 y \, dx - \frac{1}{2}(1)(1)$ $= \int_0^1 \frac{2x}{1+x^2} \, dx - \frac{1}{2}$ $= \left[\ln 1+x^2 \right]_0^1 - \frac{1}{2}$ $= \ln 2 - \frac{1}{2}$
(b)	volume of solid $= \pi \int_0^1 y^2 dx - \frac{1}{3} \pi (1)^2 (1)$ $= \pi \int_0^1 \left(\frac{2x}{1+x^2} \right)^2 dx - \frac{\pi}{3}$ $= \pi \int_0^1 \frac{4x^2}{(1+x^2)^2} dx - \frac{\pi}{3}$ $= \pi \int_0^{\frac{\pi}{4}} \frac{4 \tan^2 \theta}{(\sec^2 \theta)^2} \sec^2 \theta \, d\theta - \frac{\pi}{3}$ $= 4\pi \int_0^{\frac{\pi}{4}} \sin^2 \theta \, d\theta - \frac{\pi}{3}$ $= 4\pi \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2\theta}{2} \, d\theta - \frac{\pi}{3}$ $= 2\pi \int_0^{\frac{\pi}{4}} (1 - \cos 2\theta) \, d\theta - \frac{\pi}{3}$ $= 2\pi \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{4}} - \frac{\pi}{3}$ $= \frac{\pi^2}{2} - \frac{4\pi}{3}$

Qn	Solution
11	Maclaurin's Expansion
(a)	$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ Let $x = 1$, $e^1 = 1 + 1 + \frac{1^2}{2!} + \frac{1^3}{3!} + \dots$ $\Rightarrow e = 1 + \sum_{r=1}^{\infty} \frac{1}{r!}$ $\Rightarrow \sum_{r=1}^{\infty} \frac{1}{r!} = e - 1$
(b)(i)	$y = \tan^{-1} 2x$ $\tan y = 2x$ Diff wrt x, $\sec^2 y \frac{dy}{dx} = 2 \Rightarrow (1 + 4x^2) \left(\frac{dy}{dx} \right) = 2$ Diff wrt x, $(1 + 4x^2) \left(\frac{d^2 y}{dx^2} \right) + 8x \left(\frac{dy}{dx} \right) = 0$ Diff wrt x, $(1 + 4x^2) \left(\frac{d^3 y}{dx^3} \right) + 16x \left(\frac{d^2 y}{dx^2} \right) + 8 \left(\frac{dy}{dx} \right) = 0$ When $x = 0$, $y = 0, \frac{dy}{dx} = 2, \frac{d^2 y}{dx^2} = 0, \frac{d^3 y}{dx^3} = -16$ $\therefore y \approx 2x - \frac{8}{3}x^3$
(b)(ii)	$\int_0^{\frac{1}{5}} \tan^{-1} 2x \, dx \approx \int_0^{\frac{1}{5}} \left(2x - \frac{8}{3}x^3 \right) dx$ $= \left[x^2 - \frac{2}{3}x^4 \right]_0^{\frac{1}{5}}$ $= \frac{73}{1875}$ <u>Alternative Method</u> Using GC, and converting answer to fraction

Qn	Solution
12	Vectors 3
(i)	$3y + z = 2 \quad (1)$ $\mathbf{r} \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} = 9 \text{ in cartesian form is } x - y - 2z = 9 \quad (2)$ $2(1) + (2),$ $x + 5y = 13$ Let $x = 0$, we have $y = \frac{13}{5}$. From (1), $z = 2 - 3\left(\frac{13}{5}\right) = -\frac{29}{5}$ $\therefore C\left(0, \frac{13}{5}, -\frac{29}{5}\right)$ is another point that lies on both Π_1 and Π_2. Now $\overrightarrow{AC} = \begin{pmatrix} 0-3 \\ \frac{13}{5}-2 \\ -\frac{29}{5}+4 \end{pmatrix} = \begin{pmatrix} -3 \\ \frac{3}{5} \\ -\frac{9}{5} \end{pmatrix} = -\frac{3}{5} \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}$ $\therefore$ a direction vector for line l is $\begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}$. Hence equation of line l is $\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}, \lambda \in \mathbb{R}$. <u>Alternative Solution 1</u> Vector equation (1) in scalar product form is $\mathbf{r} \cdot \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} = 2 \quad (3)$ Vector equation (2) in scalar product form is $\mathbf{r} \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} = 9 \quad (4)$ $\mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 3 & 1 \\ 1 & -1 & -2 \end{vmatrix} = \begin{pmatrix} -5 \\ 1 \\ -3 \end{pmatrix} = -\begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}$

$\therefore$ a direction vector for line l is $\vec{m} = \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}$.

Hence equation of line l is $\vec{r} = \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}, \lambda \in \mathbb{R}$.

Alternative Solution 2

$$3y + z = 2 \quad (1)$$

$$\vec{r} \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} = 9 \text{ in cartesian form is } x - y - 2z = 9 \quad (2)$$

Using GC Simult Eqn Solver:

$$\text{Equation of line } l \text{ is } \vec{r} = \begin{pmatrix} \frac{29}{3} \\ \frac{2}{3} \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}, \lambda \in \mathbb{R}.$$

(ii) A line through B perpendicular to plane Π_2 is

$$\vec{r} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}, \alpha \in \mathbb{R}. \quad (5)$$

Subst (5) into (4):

$$\begin{pmatrix} 2 + \alpha \\ 3 - \alpha \\ -1 - 2\alpha \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} = 9$$

$$2 + \alpha - 3 + \alpha + 2 + 4\alpha = 9$$

$$\therefore \alpha = \frac{4}{3}$$

$\rightarrow$

Hence position vector $\vec{OD}$ of foot of perpendicular from B to Π_2 is

$$\vec{OD} = \begin{pmatrix} 2 + \frac{4}{3} \\ 3 - \frac{4}{3} \\ -1 - \frac{8}{3} \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 10 \\ 5 \\ -11 \end{pmatrix}$$

$$\text{Now } \vec{AD} = \frac{1}{3} \begin{pmatrix} 10 \\ 5 \\ -11 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

Hence length of projection of $\vec{AB}$ onto Π_2 is

$$\left| \vec{AD} \right| = \frac{1}{3} \sqrt{1^2 + (-1)^2 + 1^2} = \frac{\sqrt{3}}{3}.$$

Alternative Solution 1

$$\vec{AB} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$$

$$\text{Also } \vec{AD} = \frac{1}{3} \begin{pmatrix} 10 \\ 5 \\ -11 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

Hence length of projection of $\vec{AB}$ onto Π_2 is

$$\frac{\left| \vec{AB} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right|} = \frac{\left| \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right|}{\sqrt{3}} = \frac{|-1 - 1 + 3|}{\sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

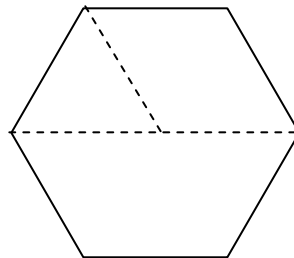
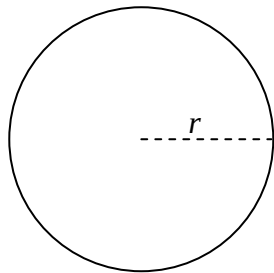
Alternative Solution 2

$$\vec{AB} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$$

	$\rightarrow$ Length of projection of AB onto Π_2 is $\left \frac{\vec{AB} \cdot \vec{n}}{ \vec{n} } \right = \left \frac{\vec{AB} \cdot \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}}{\sqrt{6}} \right $ $= \frac{\sqrt{6}}{6} \begin{vmatrix} i & j & k \\ -1 & 1 & 3 \\ 1 & -1 & -2 \end{vmatrix} = \frac{\sqrt{6}}{6} \begin{vmatrix} 1 \\ 1 \\ 0 \end{vmatrix} = \frac{\sqrt{6}}{6} (\sqrt{2}) = \frac{\sqrt{3}}{3}$
(iii)	$\vec{a} \times \vec{m} = \begin{vmatrix} i & j & k \\ 3 & 2 & -4 \\ 5 & -1 & 3 \end{vmatrix} = \begin{pmatrix} 6-4 \\ -(9+20) \\ -3-10 \end{pmatrix} = \begin{pmatrix} 2 \\ -29 \\ -13 \end{pmatrix}$ $\therefore$ vector equation of plane containing the line l and the origin is $\vec{r} \cdot \begin{pmatrix} 2 \\ -29 \\ -13 \end{pmatrix} = 0 \quad \text{or} \quad \vec{r} \cdot \begin{pmatrix} -2 \\ 29 \\ 13 \end{pmatrix} = 0$

Qn	Solution
13	Differentiation and Applications
(a)	$\frac{d}{dx} \left(\cos^{-1} \sqrt{1-4x^2} \right) = -\frac{\frac{1}{2}(1-4x^2)^{-\frac{1}{2}}(-8x)}{\sqrt{1-(1-4x^2)}}$ $= -\frac{\frac{-4x}{\sqrt{1-4x^2}}}{\sqrt{4x^2}}$ $= \frac{4x}{ 2x \sqrt{1-4x^2}}$ $= \frac{2x}{ x \sqrt{1-4x^2}}$ $= \begin{cases} \frac{2}{\sqrt{1-4x^2}} & , x \geq 0 \\ -\frac{2}{\sqrt{1-4x^2}} & , x < 0 \end{cases}$

(b)



Length of circle = $2\pi r$

Perimeter of hexagon = $d - 2\pi r$

A = total area

$$= \pi r^2 + 6 \left(\frac{1}{2} \right) \left(\frac{d - 2\pi r}{6} \right)^2 \sin \left(\frac{\pi}{3} \right)$$

$$= \pi r^2 + \frac{\sqrt{3}}{24} (d - 2\pi r)^2$$

$$\frac{dA}{dr} = 2\pi r + \frac{\sqrt{3}}{12} (d - 2\pi r) (-2\pi)$$

$$\frac{dA}{dr} = 2\pi r - \frac{\pi\sqrt{3}}{6} (d - 2\pi r)$$

$$\frac{dA}{dr} = 0 \Rightarrow 2r - \frac{\sqrt{3}}{6} (d - 2\pi r) = 0$$

$$\Rightarrow r \left(2 + \frac{\pi\sqrt{3}}{3} \right) = \frac{d\sqrt{3}}{6}$$

$$\Rightarrow r = 0.075692d$$

$$\Rightarrow r \approx 0.076d$$

$$\frac{d^2A}{dr^2} = 2\pi + \frac{\pi^2\sqrt{3}}{3} > 0$$

Thus, the sum of the areas enclosed by the two shapes is a minimum when the radius of the circle is approximately $0.076d$ units.